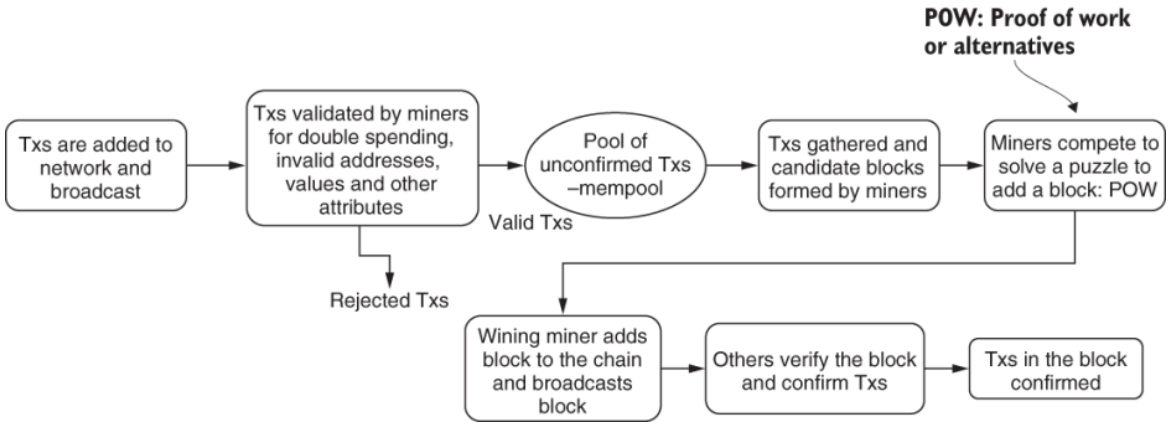


Figure 12.3 Block creation and transaction confirmation with POW consensus model

12.3.1 Proof of work

The POW algorithm has been working in Bitcoin since its inception more than ten years ago; Ethereum has used POW since its release. POW is the baseline for many consensus algorithms that are being proposed, so it is good to review POW in this section. POW works like this. Compute the hash H of block header elements (fixed) and a nonce (variable):

1. `H = hash(header, nonce) //nonce is variable parameter in the header`
2. `If H <= function(difficulty) for Ethereum, the miner has solved the puzzle, jump to step 4. //difficulty is a variable parameter in the header`
3. `Else change the nonce and repeat steps 1 and 2.`
4. `Puzzle has been solved.`

Although it is difficult to find the combination (header, nonce) that solves the problem in step 2, it is easy to verify. How do you